

JessicaGlozzo

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Languages

Italian: Native Speaker
English: Fluent

Profiles

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KQU-5176-2024

Bibliometrics

Publications: 26
Citations: 473
h-index: 10
(Scopus, Sept 2026)

Experience

01/01/2026–
Now

Translational Immunology Unit, Fondazione IRCCS Istituto Nazionale dei Tumori Milan, Italy
Bioinformatician (institutional fellowship)

- Analysis of multi-omics longitudinal data from blood samples of breast cancer patients to evaluate the effect of a fasting-mimicking diet on immunity.
- Support wet-lab team members in bioinformatics analysis (mostly RNA-sequencing downstream analysis), including experimental design, data interpretation, and hypothesis generation.
- Development of an R package for the analysis of longitudinal data.
- Ongoing collaboration with the BiomiX consortium to evaluate updated multi-omics integration pipelines and contribute to a research manuscript.
- Ongoing collaboration with the University of Milan to investigate the problem of selection bias in GCN-based multi-omics integration pipelines.

Initial engagement began through short-term and collaborative roles within the same unit.

01/02/2024–
31/08/2025

Department of Computer Science, Università degli Studi di Milano Milan, Italy
Postdoctoral Research Fellow

Development and application of methods for the integration and analysis of heterogeneous biological data (genomic, epigenomic, transcriptomic, proteomic and clinical data) to support precision medicine and to improve disease prognosis, diagnosis and treatment. I mainly used approaches based on Patient Similarity Networks (PSN) and graph analysis, that enable the integration of high-dimensional multimodal data obtaining state-of-the-art results in patient subtyping and clinical outcome prediction. I worked on the following research activities:

- Experimental validation, manuscript writing, and preparation of responses to reviewers for the paper proposing miss-SNF (DOI: 10.1093/bioinformatics/btaf150), a novel PSN-based data-fusion approach designed to integrate multimodal biomedical data with missing modalities. miss-SNF employs a nonlinear message-passing strategy to integrate incomplete datasets and partially reconstruct missing similarities using patient neighborhood information, enabling both unsupervised and (semi-) supervised analyses.
- Unsupervised analysis of multimodal data (laboratory measurements, genetic mutations, cytometric markers, fusion transcript expression, and longitudinal data) from Acute Myeloid Leukemia patients to identify clinically relevant patient subgroups. The analysis leveraged PSN-based methods for data integration, spectral clustering, and downstream validation approaches, including survival analysis and clinical variable enrichment.

- Integration of PSN-based data-fusion methods (i.e., SNF and NEMO), together with clustering, validation, analysis, and visualization techniques of the identified clusters, into BiomiX –a state-of-the-art bioinformatics tool that makes multi-omics analysis methods accessible to researchers without specific expertise in the field. This work is carried out within the “BiomiX consortium”, which includes several research groups (LBAI - B Lymphocytes, Autoimmunity and Immunotherapies - Inserm U1227 - Université de Bretagne Occidentale; University of Granada; Universidad CEU San Pablo; UMR1078 GGB - Université de Bretagne Occidentale; University of Zurich).
- Collaboration with the Molecular Oncology group at Universidad Francisco de Vitoria (Madrid) to analyze pancreatic cancer multi-omics data aimed at identifying molecular subtypes and prognostic signatures, using both PSN-based integration methods (such as miss-SNF and NEMO) and matrix factorization-based approaches, including MOFA and intNMF.
- Co-supervision of two Master’s theses on PSN-based multimodal integration and Graph Neural Networks, including the development of an SNF-GraphSAGE framework for molecular subtype classification and the assessment of selection bias in a GCN-based multi-omics integration method for patient classification.

01/11/2020–
31/01/2024

**Department of Computer Science, Università degli Studi di Milano;
Joint Research Centre (JRC), DG European Commission**

Italy

PhD student

PhD within the framework of the Collaborative Doctoral Partnership programme between the University of Milan and the Joint Research Centre (JRC), Directorate-General of the European Commission (Ispra, VA).

The PhD was completed on 17/04/2024 with the defense of the dissertation entitled “Patient similarity networks-based methods for multimodal data integration and clinical outcome prediction” (cum laude), which focused on:

- An innovative literature review on multimodal integration approaches for the construction of patient similarity networks (PSN).
- The development of a multimodal data-fusion pipeline leveraging block-wise analysis to adaptively reduce data dimensionality through the estimation of intrinsic dimensionality.
- The development of miss-SNF, a method for the integration of multimodal data in the presence of patients with completely missing modalities. This approach performs nonlinear integration of unimodal PSN, enables partial reconstruction of missing similarities within the integrated network, and supports both (semi-) supervised and unsupervised prediction tasks.

Main publications resulting from the research activities carried out during the PhD:

- Briefings in Bioinformatics (2022), DOI: 10.1093/bib/bbac207
- Artificial Intelligence in Medicine (2025), DOI: 10.1016/j.artmed.2024.103049
- Bioinformatics (2025), DOI: 10.1093/bioinformatics/btaf150

The dissertation is available at: <https://hdl.handle.net/20.500.14242/85308>.

01/02/2020–
31/10/2020

Neuroradiology Unit, IRCCS Ospedale San Raffaele

Milan, Italy

Researcher

Application of statistical methods to decompose multi-exponential T2 signals from brain magnetic resonance imaging data, with the aim of deriving myelin water fraction maps and investigating white matter development in healthy pediatric subjects and patients affected by metachromatic leukodystrophy.

The main outcome was the implementation of a software pipeline for:

- computation of myelin water fraction maps;
- automated brain segmentation using a U-Net convolutional neural network;
- co-registration with the Myelin Water Atlas;
- statistical analysis in the regions of interest.

01/03/2017–
31/01/2020

Dermatology Unit, Fondazione IRCCS Ca' Granda - Ospedale Maggiore Policlinico Milan, Italy
Bioinformatician (institutional fellowship)
Bioinformatic analysis of DNA and RNA sequencing data generated by Next Generation Sequencing technologies (Illumina platform), aimed at identifying diagnostic, prognostic, and therapeutic biomarkers in patients affected by blastic plasmacytoid dendritic cell neoplasm and aggressive epidermotropic primary cutaneous CD8 T-cell lymphoma.
The project involved the design and implementation of bioinformatics workflows. RNA-seq analyses included differential expression profiling between normal and tumor samples, downstream functional enrichment analyses, and result visualization. DNA-seq analyses focused on the identification of somatic variants and structural alterations through end-to-end processing from raw sequencing data to annotated results.

Visiting researcher

16-30/10/2018
15-30/11/2019

KIRCHER LAB - Computational Genome Biology Group, Berlin Institute of Health, Charité Berlin, Germany
Visiting researcher
Research activity within the MIUR-DAAD Joint Mobility Program-funded project “Developing machine learning methods for the prioritization of regulatory variants in human disease”, promoting collaboration between Italian and German research groups.

01/03/2016–
30/06/2016

PaccanaroLab, Centre for Systems Biology, Department of Computer Science - Royal Holloway, University of London Egham, TW20 0EX, UK
Visiting researcher
International research collaboration with the PaccanaroLab to conduct part of my Master's thesis, supported by funding from the University of Milan.

Selected scientific projects

01/07/2024–
30/06/2025

IntPartPSN: Multi-modal data INTEGRation of PARTial Patient Similarity Networks for precision medicine tasks
Principal Investigator
Development of the miss-SNF method and investigation of selection bias in a GCN-based approach for multi-omics integration and patient classification. This project was funded with 25000 computing hours on INDACO HPC (University of Milan HPC).

2018–2020

Developing machine learning methods for the prioritization of regulatory variants in human diseases
Collaborator
Within this broader project, I contributed to the prediction of the activity status of tissue-specific cis-regulatory regions (i.e. promoters and enhancers) in the human genome, using feed-forward neural networks exploiting epigenomic data and convolutional neural networks trained on sequence data. The project was a collaboration between Università degli Studi di Milano and Berlin Institute of Health (BIH), funded by the German DAAD and Italian MIUR in the “MIUR-DAAD Joint Mobility Program”.

2020–2021

Multicriteria Data Structures and Algorithms: from compressed to learned indexes, and beyond
Collaborator (This project funded my first year of PhD)
I developed a case study demonstrating that applying compression techniques to pre-trained neural networks for cell counting in digital pathology is an effective strategy to reduce computational resource consumption (i.e., disk space and memory) and energy usage, while preserving—or even improving—model accuracy. The resulting compact models can be deployed on devices with limited resources (e.g., smartphones, tablets, or older computers).

2022–2024

Automated Viral Genome Analysis for SARS-CoV-2 Genomic Surveillance (2022-2024)

Collaborator

Collaboration with the Department of Biomedical and Clinical Sciences, Università degli Studi di Milano to identify and evaluate automated pipelines for the analysis of SARS-CoV-2 viral genome sequencing data, aimed at monitoring and characterizing viral lineages through the identification of mutations of biological and public health relevance.

Education

01/11/2020–
17/04/2024

PhD in Computer Science

Università degli Studi di Milano, Milan, Italy

Collaborative Doctoral Partnership (CDP) programme between Università degli Studi di Milano and Joint Research Centre, European Commission.

Final evaluation: Cum laude (thesis defense: 17 April 2024)

Doctoral thesis: "Patient similarity networks-based methods for multimodal data integration and clinical outcome prediction", Supervisor: Prof. Giorgio Valentini, Co-supervisors: Prof. Elena Casiraghi and Dr. Alex Patak.

12/03/2014–
12/12/2016

Master's Degree in Molecular Biotechnology and Bioinformatics Università degli Studi di Milano, Milan, Italy

A portion of the thesis research was carried out at the PaccanaroLab, Centre for Systems Biology, Department of Computer Science – Royal Holloway, University of London, supported by a scholarship from the University of Milan. The work was later expanded and published in Scientific Reports (Nature Portfolio) (DOI: 10.1038/s41598-020-60235-8).

Final evaluation: 110/110 with honors (cum laude) (thesis defense: 12 December 2016)

Thesis: "Network-based methods for outcome prediction in the 'sample space' ", Supervisor: Prof. Giorgio Valentini, Co-supervisors: Prof. Alberto Paccanaro and Prof. Matteo Re.

16/09/2010–
21/02/2014

Bachelor's Degree in Medical Biotechnology

Università degli Studi di Milano, Milan, Italy

Final evaluation: 104/110 (thesis defense: 21 February 2014)

Thesis: "Comparazione tra approccio biochimico e imaging con CCD camera per la valutazione dell'attività luciferasica in colture cellulari" (Comparison between biochemical assay and CCD camera imaging for the evaluation of luciferase activity in cell cultures), Supervisor: Prof. Luisa Ottobrini.

09/2005–
07/2010

High School Diploma - Scientific and Technological Curriculum (Liceo Scientifico Tecnologico)

ITIS "Augusto Righi", Corsico (MI), Italy

Final evaluation: 100/100

Other training activities

24/07/2022–
30/07/2022

Attendance at the Summer School "Jacob T. Schwartz International School for Scientific Research - Lipari School on Computational Life Sciences" entitled "Artificial Intelligence in Biomedicine"

Jacob T. Schwartz International School for Scientific Research, Università degli Studi di Catania

16/07/2018–
19/07/2018

Attendance at the course "VIII Edition - NGS data analysis and applications in diagnostic"

University of Pavia

Skills

Computational and analytical expertise:

1. Multimodal/multi-omics data integration
2. Patient similarity networks and graph-based modeling
3. Supervised and unsupervised machine learning
4. Deep learning applications for biological data
5. Feature selection and dimensionality reduction
6. RNA-seq and whole-genome sequencing data analysis
7. Exploratory data analysis and visualization

Programming and Infrastructure:

1. R (advanced) – statistical modeling, multi-omics workflows, development and maintenance of open-source R packages
2. Python
3. Linux-based Operating Systems and terminal
4. Linux-based systems administration (user provisioning, SSH key-based authentication)
5. HPC environments (Slurm workload manager)
6. Git/GitHub
7. Conda environments
8. Agentic coding (OpenAI Codex CLI)

Research and Scientific Skills:

- Reproducible computational research practices (version control, environment management, controlled randomness)
- Design and validation of computational experiments
- Technical report and manuscript writing
- Scientific communication (lectures, seminars, conferences)
- Co-supervision of MSc and BSc theses
- Multidisciplinary (computer scientists, wet-lab researchers, clinicians) and international collaboration

Design, development, and maintenance of software libraries

All the following software libraries are open-access to support the scientific community.

- Developer of *miss-SNF*: a multimodal patient similarity network integration approach to handle completely missing data sources.
R package available at: <https://github.com/GliozzoJ/missSNF>.
- Developer of *P-Net* (Patient-Net): Network-based ranking of patients with respect to a given phenotype/outcome.
R package available at: <https://github.com/GliozzoJ/P-Net>.
- Developer of *pathonet_compression*: Reducing the Complexity of Deep Learning Models for Medical Applications in Resource-limited Contexts.
Python repository available at: https://github.com/GliozzoJ/pathonet_compression.
- Collaborator to the software *PathoNet*: deep neural network backend for evaluation of Ki-67 and tumor-infiltrating lymphocytes in breast cancer.
Python repository available at: <https://github.com/SHIDCenter/PathoNet>.
- Maintainer of *NetInt*: Methods for Unweighted and Weighted Network Integration.
R package available on CRAN: <https://cran.r-project.org/web/packages/NetInt/index.html>.

- Maintainer of *clusterv*: Assessment of Cluster Stability by Randomized Maps.
R package available on CRAN: <https://cran.r-project.org/web/packages/clusterv/index.html>.
- Maintainer of *mosclust*: Model Order Selection for Clustering.
R package available on CRAN: <https://cran.r-project.org/web/packages/mosclust/index.html>.

Teaching and Mentoring

2022–2024 **Teaching Assistant, "Bioinformatics" course (MSc in Computer Science), A.Y. 2022-2023 and 2023-2024** Università degli Studi di Milano, Milan, Italy

Organized and delivered laboratory sessions and final practical part of the exam (18 hours total).

Topics: Classification and disease subtype discovery (clustering) using single and multi-omics data.

2020-2025 **Guest Lecturer** Università degli Studi di Milano, Milan, Italy

Guest lecturer in the following courses:

- "Filogenesi applicata alla epidemiologia e sorveglianza genomica degli agenti infettivi" (Applied Phylogenetics for Epidemiology and Genomic Surveillance of Infectious Agents) organized by the Department of Biomedical and Clinical Sciences in the 2024 and 2025 editions.
- "Bioinformatics" (MSc in Computer Science) in the A.Y. 2020-2021 and 2021-2022.
- "Principles and Models of Perception" (BSc in Computer Science for New Media Communications) in A.Y. 2021-2022.

Topics: Kernels on graphs and the P-Net algorithm; Automated pipelines for SARS-CoV-2 viral genome analysis.

2020–2025 **Co-supervisor of 4 BSc and 4 MSc theses.** Università degli Studi di Milano, Milan, Italy

Topics: PSN construction, multimodal integration, graph neural networks, viral genomics and machine learning for biomedical data.

Conferences, Workshops and Tutorials

- Speaker at the conference "BIOINFORMATICS 2023 - 14th International Conference on Bioinformatics Models, Methods and Algorithms" (Lisbon, Portugal) on 16-18 February 2023. Presentation topic: "Patient Similarity Networks Integration for Partial Multimodal Datasets".
- Co-instructor of a tutorial at the 25th European Conference on Computational Biology (ECCB 2026 - 31 August to 4 September 2026, Geneva, Switzerland) - Tutorial title: Democratizing multi-omics data analysis with BiomiX and Nextflow. Material available at this link.

Scientific service

- Program Committee Member for the workshop "SAI-Care 2024: Integrating Symbiotic AI in Biomedical Informatics, A Workshop on Advanced SAI-based applications in Healthcare", part of the IEEE BIBM 2024 conference.
- Editorial Board Member of the journal BMC Artificial Intelligence.
- Guest Editor for Discover Computing (Springer Nature) - Collection Proposal: Graph Representation Learning for biomedical applications.
- Guest Editor for Frontiers in Medicine - Research Topic: Multi-Omics Integration for Clinical Decision-Making: Methods, Models, and Applications.
- Community Reviewer for Frontiers in Genetics - Computational Genomics.
- Reviewer for international journals including BMC Bioinformatics, Nature Communications, Women's Health - Nature Portfolio journals, Scientific Reports - Nature Portfolio journals, International Journal of Machine Learning and Cybernetics, Digital Medicine - Nature Portfolio

journals, Applied Network Science, BMC Medical Informatics and Decision Making, Journal of Big Data, Communications Medicine - Nature Publishing Group.

- Member of BITS - Società Italiana di Bioinformatica (Italian Society of Bioinformatics).

Publications

Peer-reviewed journals

- [R1] Francesco Torgano, Mauricio Soto Gomez, Matteo Zignani, **Jessica Gliozzo**, Emanuele Cavalleri, Marco Mesiti, Elena Casiraghi and Giorgio Valentini. "RNA knowledge-graph analysis through homogeneous embedding methods". In: *Bioinformatics Advances* 5.1 (13th May 2025), vbaf109. ISSN: 2635-0041. DOI: 10.1093/bioadv/vbaf109. eprint: <https://academic.oup.com/bioinformaticsadvances/article-pdf/5/1/vbaf109/63166447/vbaf109.pdf>. URL: <https://doi.org/10.1093/bioadv/vbaf109>.
- [R2] **Jessica Gliozzo**, Mauricio A Soto Gomez, Arturo Bonometti, Alex Patak, Elena Casiraghi and Giorgio Valentini. "miss-SNF: a multimodal patient similarity network integration approach to handle completely missing data sources". In: *Bioinformatics* 41.4 (4th April 2025), btaf150. ISSN: 1367-4811. DOI: 10.1093/bioinformatics/btaf150. eprint: <https://academic.oup.com/bioinformatics/article-pdf/41/4/btaf150/62871738/btaf150.pdf>. URL: <https://doi.org/10.1093/bioinformatics/btaf150>.
- [R3] **Jessica Gliozzo**, Mauricio Soto-Gomez, Valentina Guarino, Arturo Bonometti, Alberto Cabri, Emanuele Cavalleri, Justin Reese, Peter N. Robinson, Marco Mesiti, Giorgio Valentini and Elena Casiraghi. "Intrinsic-dimension analysis for guiding dimensionality reduction and data fusion in multi-omics data processing". In: *Artificial Intelligence in Medicine* 160 (February 2025), p. 103049. ISSN: 0933-3657. DOI: 10.1016/j.artmed.2024.103049. URL: <https://doi.org/10.1016/j.artmed.2024.103049>.
- [R4] Francesco Torgano, Emanuele Cavalleri, **Jessica Gliozzo**, Federico Stacchiotti, Emanuele Saitto, Marco Mesiti, Elena Casiraghi and Giorgio Valentini. "RNA Knowledge Graph Analysis via Embedding Methods". In: *WSEAS Transactions on Biology and Biomedicine* 21 (2024), pp. 302–312. ISSN: 1109-9518. DOI: 10.37394/23208.2024.21.30. URL: [https://wseas.com/journals/bab/2024/a605108-022\(2024\).pdf](https://wseas.com/journals/bab/2024/a605108-022(2024).pdf).
- [R5] Emanuele Cavalleri, Alberto Cabri, Mauricio Soto-Gomez, Sara Bonfitto, Paolo Perlasca, **Jessica Gliozzo**, Tiffany J Callahan, Justin Reese, Peter N Robinson, Elena Casiraghi, Giorgio Valentini and Marco Mesiti. "An ontology-based knowledge graph for representing interactions involving RNA molecules". In: *Scientific Data* 11.1 (22nd August 2024), p. 906. ISSN: 2052-4463. DOI: 10.1038/s41597-024-03673-7. URL: <https://doi.org/10.1038/s41597-024-03673-7>.
- [R6] Giorgio Valentini, Dario Malchiodi, **Jessica Gliozzo**, Marco Mesiti, Mauricio Soto-Gomez, Alberto Cabri, Justin Reese, Elena Casiraghi and Peter N. Robinson. "The promises of large language models for protein design and modeling". In: *Frontiers in Bioinformatics* 3 (23rd November 2023), p. 1304099. ISSN: 2673-7647. DOI: 10.3389/fbinf.2023.1304099. URL: <https://www.frontiersin.org/articles/10.3389/fbinf.2023.1304099>.
- [R7] Luca Cappelletti, Alessandro Petrini, **Jessica Gliozzo**, Elena Casiraghi, Max Schubach, Martin Kircher and Giorgio Valentini. "Boosting tissue-specific prediction of active cis-regulatory regions through deep learning and Bayesian optimization techniques". In: *BMC Bioinformatics* 23.2 (12th December 2022), p. 154. ISSN: 1471-2105. DOI: 10.1186/s12859-022-04582-5. URL: <https://doi.org/10.1186/s12859-022-04582-5>.
- [R8] **Jessica Gliozzo**, Marco Mesiti, Marco Notaro, Alessandro Petrini, Alex Patak, Antonio Puertas-Gallardo, Alberto Paccanaro, Giorgio Valentini and Elena Casiraghi. "Heterogeneous data integration methods for patient similarity networks". In: *Briefings in Bioinformatics* 23.4 (13th June 2022), bbac207. ISSN: 1477-4054. DOI: 10.1093/bib/bbac207. eprint: <https://academic.oup.com/bib/article-pdf/23/4/bbac207/45016588/bbac207.pdf>. URL: <https://doi.org/10.1093/bib/bbac207>.
- [R9] Arturo Bonometti, **Jessica Gliozzo**, Chiara Moltrasio, Filippo Bagnoli and Emilio Berti. "Cutaneous-group histiocytoses associated with myeloid malignancies: A systematic review of 102 cases". In: *Australasian Journal of Dermatology* 62.2 (2021), e162–e169. ISSN: 0004-8380. DOI: <https://doi.org/10.1111/ajd.13491>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/ajd.13491>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/ajd.13491>.

- [R10] Marco Notaro, Marco Frasca, Alessandro Petrini, **Jessica Gliozzo**, Elena Casiraghi, Peter N Robinson and Giorgio Valentini. "HEMDAG: a family of modular and scalable hierarchical ensemble methods to improve Gene Ontology term prediction". In: *Bioinformatics* 37.23 (December 2021), pp. 4526–4533. ISSN: 1367-4803. DOI: [10.1093/bioinformatics/btab485](https://doi.org/10.1093/bioinformatics/btab485). eprint: <https://academic.oup.com/bioinformatics/article-pdf/37/23/4526/50579686/btab485.pdf>. URL: <https://doi.org/10.1093/bioinformatics/btab485>.
- [R11] Adriana Cassaro, Giovanni Grillo, Marco Notaro, **Jessica Gliozzo**, Ilaria Esposito, Gianluigi Reda, Alessandra Trojani, Giorgio Valentini, Barbara Di Camillo, Roberto Cairoli and Alessandro Beghini. "FZD6 triggers Wnt–signalling driven by WNT10B/VS1 expression and highlights new targets in T-cell acute lymphoblastic leukemia". In: *Hematological Oncology* 39.3 (August 2021), pp. 364–379. ISSN: 0278-0232. DOI: <https://doi.org/10.1002/hon.2840>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/hon.2840>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/hon.2840>.
- [R12] Paolo Perlasca, Marco Frasca, Cheick Tidiane Ba, **Jessica Gliozzo**, Marco Notaro, Mario Pennacchioni, Giorgio Valentini and Marco Mesiti. "Multi-resolution visualization and analysis of biomolecular networks through hierarchical community detection and web-based graphical tools". In: *PLOS ONE* 15.12 (22nd December 2020), e0244241. ISSN: 1932-6203. DOI: [10.1371/journal.pone.0244241](https://doi.org/10.1371/journal.pone.0244241). URL: <https://doi.org/10.1371/journal.pone.0244241>.
- [R13] Armando N. Bastidas Torres, Davy Cats, Hailiang Mei, Daniele Fanoni, **Jessica Gliozzo**, Laura Corti, Marco Paulli, Maarten H. Vermeer, Rein Willemze, Emilio Berti and Cornelis P. Tensen. "Whole-genome analysis uncovers recurrent IKZF1 inactivation and aberrant cell adhesion in blastic plasmacytoid dendritic cell neoplasm". In: *Genes, Chromosomes and Cancer* 59.5 (May 2020), pp. 295–308. ISSN: 1045-2257. DOI: <https://doi.org/10.1002/gcc.22831>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/gcc.22831>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/gcc.22831>.
- [R14] **Jessica Gliozzo**, Paolo Perlasca, Marco Mesiti, Elena Casiraghi, Viviana Vallacchi, Elisabetta Vergani, Marco Frasca, Giuliano Grossi, Alessandro Petrini, Matteo Re, Alberto Paccanaro and Giorgio Valentini. "Network modeling of patients' biomolecular profiles for clinical phenotype/outcome prediction". In: *Scientific Reports* 10.1 (27th February 2020), p. 3612. ISSN: 2045-2322. DOI: [10.1038/s41598-020-60235-8](https://doi.org/10.1038/s41598-020-60235-8). URL: <https://doi.org/10.1038/s41598-020-60235-8>.
- [R15] Barbara Rita Barricelli, Elena Casiraghi, **Jessica Gliozzo**, Alessandro Petrini and Stefano Valtolina. "Human Digital Twin for Fitness Management". In: *IEEE Access* 8 (4th February 2020), pp. 26637–26664. ISSN: 2169-3536. DOI: [10.1109/ACCESS.2020.2971576](https://doi.org/10.1109/ACCESS.2020.2971576). URL: <https://doi.org/10.1109/ACCESS.2020.2971576>.
- [R16] Barbara Rita Barricelli, Elena Casiraghi, **Jessica Gliozzo**, Veronica Huber, Biagio Eugenio Leone, Alessandro Rizzi and Barbara Vergani. "ki67 nuclei detection and ki67-index estimation: a novel automatic approach based on human vision modeling". In: *BMC Bioinformatics* 20.1 (27th December 2019), p. 733. ISSN: 1471-2105. DOI: [10.1186/s12859-019-3285-4](https://doi.org/10.1186/s12859-019-3285-4). URL: <https://doi.org/10.1186/s12859-019-3285-4>.
- [R17] Paolo Perlasca, Marco Frasca, Cheick Tidiane Ba, Marco Notaro, Alessandro Petrini, Elena Casiraghi, Giuliano Grossi, **Jessica Gliozzo**, Giorgio Valentini and Marco Mesiti. "UNIPred-Web: a web tool for the integration and visualization of biomolecular networks for protein function prediction". In: *BMC Bioinformatics* 20.1 (14th August 2019), p. 422. ISSN: 1471-2105. DOI: [10.1186/s12859-019-2959-2](https://doi.org/10.1186/s12859-019-2959-2). URL: <https://doi.org/10.1186/s12859-019-2959-2>.
- [R18] Luca Cappelletti, **Jessica Gliozzo**, Alessandro Petrini and Giorgio Valentini. "Training Neural Networks with Balanced Mini-batch to Improve the Prediction of Pathogenic Genomic Variants in Mendelian Diseases". In: *Sensors & Transducers* 234.6 (30th June 2019), pp. 16–21. ISSN: 2306-8515. URL: https://sensorsportal.com/p_3087.html.
- [R19] Arturo Bonometti, **Jessica Gliozzo**, Chiara Moltrasio, Filippo Bagnoli, Luigia Venegoni, Emanuela Passoni, Marco Paulli and Emilio Berti. "Disfiguring Nodular Cephalic Xanthoma Disseminatum: An Exceptional Variant of a Forgotten Entity". In: *Acta Dermato-Venereologica* 99.4 (April 2019), pp. 450–451. ISSN: 1651-2057. DOI: [10.2340/00015555-3111](https://doi.org/10.2340/00015555-3111). URL: <https://medicaljournalssweden.se/actadv/article/view/3161>.
- [R20] Marco Frasca, Giuliano Grossi, **Jessica Gliozzo**, Marco Mesiti, Marco Notaro, Paolo Perlasca, Alessandro Petrini and Giorgio Valentini. "A GPU-based algorithm for fast node label learning in large and unbalanced biomolecular networks". In: *BMC Bioinformatics* 19.10 (15th October 2018), p. 353. ISSN: 1471-2105. DOI: [10.1186/s12859-018-2301-4](https://doi.org/10.1186/s12859-018-2301-4). URL: <https://doi.org/10.1186/s12859-018-2301-4>.

- [PP1] Laura Bergamaschi, Stefano Percio, Yiyi Zhu, Gabriele Tine, Rosalba Miceli, Marco Fiore, Elena Palassini, Paola Collini, Federica Perrone, Francesca Rini, **Jessica Gliozzo**, Cristina Banfi, Barbara Vergani, Biagio Eugenio Leone, Armando Giuseppe Licata, Loris De Cecco, Monica Zucchini, Arabella Mazzocchi, Sandro Pasquali, Alessandro Gronchi, Licia Rivoltini, Viviana Vallacchi and Chiara Colombo. "Active Surveillance Reveals a Systemic Pro-Resolving Th2 Immune Program Linked to Desmoid Tumor Regression". In: *bioRxiv* (2026). [Preprint]. doi: 10.64898/2026.04.16.718860. eprint: <https://www.biorxiv.org/content/early/2026/04/20/2026.04.16.718860.full.pdf>. URL: <https://www.biorxiv.org/content/early/2026/04/20/2026.04.16.718860>.

Conference proceedings

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